



OPEN Deep bayesian neural networks for UWB phase error correction in positioning systems

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Angle-based positioning systems have emerged as critical technologies for precise indoor localization across robotics, healthcare, and industrial automation applications. Ultrawideband (UWB) phase-based angle measurements offers theoretical sub-degree accuracy, but practical implementations suffer from channel inconsistency errors that significantly degrade performance. A dual-layer Bayesian neural network fusion framework (DBNNFF) was presented that effectively addresses these systematic errors through an innovative combination of physical constraints and uncertainty-aware modeling. Experiments were conducted in a microwave anechoic chamber using a customized 5-channel UWB base station and single-channel tags. Data was collected across seven azimuth angles between $\pm 30^\circ$, with 30s cold-start cycles per angle. The DBNNFF framework reduced the angle errors by 94.7% to $0.1036^\circ \pm 0.0182^\circ$, outperforming many existing algorithms by 25–42.1%. The framework's dual-network architecture—combining channel correlation model and cold start state distribution estimator—with uncertainty-weighted Bayesian fusion provides well-calibrated confidence intervals and exceptional noise robustness. Experiments conducted in multi-path environments such as office and hallway demonstrated that the DBNNFF algorithm exhibited robust performance, with errors maintained within 0.17° .

Ultrawideband (UWB) is characterized by nanosecond-scale pulses, an exceptionally wide spectrum and centimeter-level positioning accuracy¹. It has been extensively deployed in high-precision indoor location scenarios, including Industry 4.0, augmented-reality applications and healthcare monitoring². Phase difference-of-arrival (PDOA) techniques estimate the direction of incidence by exploiting phase disparities among multiple antennas^{3–5}. When combined with two-way ranging, they enable single-anchor localization. In comparison with the extensively adopted time-difference-of-arrival (TDOA) method^{6,7}, this approach demands reduced synchronization requirements, augmented architectural flexibility and enhanced resilience to environmental dynamics.

UWB radio-frequency front ends have clear channel mismatches after each power-on cycle. This leads to a consistent cold start error^{8,9}. The issue worsens with the common DW1000 UWB chipset. Its low-power design depends on limited resolution quantization, which uses 1-bit to 6-bit ADCs¹⁰. Without correction, these biases shift the mean AOA estimate by 3–4° and greatly inflate its variance. This constrains the attainable accuracy of single-anchor UWB localization systems^{11,12}.

Existing calibration strategies can be categorized into three broad classifications. The first method is hardware-level calibration, wherein channel errors are measured offline with a precision vector network analyzer or auxiliary calibration board and subsequently compensated via look-up tables. In this context, prior work proposed two methodologies for the amplitude-phase analysis of channels via vector network analyzers¹³. The second category comprises analytical algorithms, such as the least-squares method (LSM) and extended Kalman filtering (EKF) which assume the errors exhibit linear or weakly nonlinear dependence on angle or time. In the field of analysis algorithms, the extended Kalman filtering method has been proposed by certain researchers for the purpose of adapting estimates of the channel amplitude and phase parameters¹⁴. The maximum likelihood estimation method has been employed by other researchers to jointly estimate the I/Q mismatch, the dc offset, the carrier frequency offset and the channel for receivers in frequency-selective channels¹⁵. The third method is data-driven techniques, including Gaussian process regression (GPR) and deep learning (DL), which approximate the complex mapping directly from data. For such methods, researchers have proposed a new deep

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convolutional neural network model for the analysis of the relationship between multiple channels and the correction of data^{16,17}.

However, hardware-level approaches are costly, fail to construct explicit error models and lack online adaptability. LSM/EKF-based models struggle to capture the strong nonlinear coupling among multiple antennas, resulting in Underfitting. Conventional DL/GPR methods, despite their nonlinear expressiveness, seldom incorporate physical priority of radio interferometry, resulting in limited interpretability and extrapolation capability. In addition, there are few studies in the existing literature that specifically address multi-channel AOA measurements^{18,19}.

To surmount this impediment, the DBNNFF algorithm is proposed, which accomplishes collaborative error suppression through three innovations. First, a channel calibration module (CCM) is employed for the purpose of modelling channel correlations, in conjunction with a cold start state distribution estimator (CDE) for operational state estimation. This results in the creation of a dual-path Bayesian architecture that captures spatial correlations via channel attention mechanisms and state uncertainties through MC Dropout²⁰. Interferometer angle measurement physical constraints are explicitly incorporated in the loss function²¹, ensuring that predictions adhere to fundamental phase-angle relationships. Second, a dynamic fusion mechanism is designed since uncertainty, thereby enabling complementary advantages to be realized between the CCM and the CDE. Furthermore, to address the issue of phase ambiguity in angle measurement experimental data which has an impact on the training of Bayesian networks, the K-means clustering method, with adaptive rules to eliminate ambiguity, has been employed²². Across 12,600 real measurements from 30 cycles, DBNNFF reduced MAE to 0.13° and achieved over 95% confidence interval coverage. The study validates physically constrained Bayesian models and provides a general method for error modeling and reliability assessment in other phase measurement systems.

Methods

Problem description

UWB multi-channel AOA estimation modeling

It is imperative to consider a linear array that contains M antennas, with array element spacing d , carrier frequency f_c , and wavelength $\lambda = c/f_c$. The phase of the incident signal transmitted to the m th antenna in the single-path near-field case can be expressed as follows²¹:

$$\phi_m = \frac{2\pi d(m-1)}{\lambda} \sin \theta + \psi_m + n_m \quad (1)$$

In this equation, θ represents the azimuth angle that is being estimated, ψ_m denotes the hardware-dependent fixed phase deviation, and $n_m \sim \mathcal{N}(0, \sigma_n^2)$ indicates the measurement noise.

The observable measurement is typically the phase difference between adjacent antennas.

$$\Delta\phi_m = \phi_{m+1} - \phi_m = \frac{2\pi d}{\lambda} \sin \theta + \epsilon_m \quad (2)$$

where $\epsilon_m = \psi_{m+1} - \psi_m + n_{m+1} - n_m$.

The mathematical representation of the channel mismatch error

The UWB module has been observed to generate a one-time random response drift in its RF link after each cold start, denoted as $\delta_m^{(b)}$ (the b -th cold start). The effective phase difference at the b -th cold start is subsequently determined.

$$\Delta\phi_m^{(b)} = \frac{2\pi d}{\lambda} \sin \theta + \epsilon_m + \delta_m^{(b)} \quad (3)$$

It is evident that $\delta_m^{(b)}$ is approximately stable during the same run, however, it is independently distributed across cold start events. Furthermore, there may be a correlation in the measurement errors between different antenna pairs.

Formal definition of the cold start error modeling problem

The existence of a dataset is assumed.

$$\mathcal{D} = \left\{ \left(\mathbf{x}_i^{(b)}, \theta_i^{(b)} \right) \right\}_{i=1}^N, \quad \mathbf{x}_i^{(b)} = \left[\Delta\phi_1^{(b)}, \dots, \Delta\phi_{M-1}^{(b)} \right]^\top \quad (4)$$

where $b \in \{1, \dots, B\}$ is used to denote the number of cold starts, to learn a conditional distribution.

$$p(\theta, \sigma^2 | \mathbf{x}; \mathcal{M}) \quad (5)$$

First, its mean $\hat{\theta} = f_\mu(\mathbf{x})$ approximates the true angle. Second, its variance $\hat{\sigma}^2 = f_\sigma(\mathbf{x})$ can serve as a reliable metric for quantifying the goniometric uncertainty. Third, the model demonstrates a high degree of generalizability across each cold start b . This property enables the removal of systematic biases that may be present due to $\delta_m^{(b)}$. The DBNNFF algorithm proposed in this paper attains these objectives by integrating physically constrained regular terms with the Bayesian inference of the deep ensemble.

DBNNFF algorithm architecture

The DBNNFF algorithm is an innovative framework for modeling cold start error and channel mismatch errors in UWB multi-channel phase-based angle measurements. The overall architecture of the algorithm is illustrated in Fig. 1. It integrates five core modules: data preprocessing, K-means clustering phase unwrapping, physical constraints, dual-layer neural networks, and Bayesian fusion. The framework processes raw UWB phase data and feeds the features into two complementary networks: the CCM and the CDE, which implement uncertainty estimation through MC Dropout. Predictions are combined via uncertainty-based adaptive weighting, generating angle error estimates with confidence intervals. The physical constraint module ensures predictions conform to interferometric principles, enhancing interpretability and generalization capability across diverse conditions.

K-means clustering-based phase unwrapping

Phase cluster segmentation

Given the phase-difference observation sequence acquired in one cold start test as

$$\{\Delta\phi_1, \Delta\phi_2, \dots, \Delta\phi_n\} \subset (-\pi, \pi] \quad (6)$$

A mixture originating from K fixed angles of incidence $\{\theta_1, \dots, \theta_K\}$ was assumed, each maintaining short quasi-stationary segments in the time domain. To isolate the phase jumps caused by the switching angles, the observations are first clustered²².

(1) Data representation.

Because the conventional Euclidean distance cannot cope with the periodicity of angles, each $\Delta\phi_i$ is mapped to a 2-D unit-circle coordinate.

$$\mathbf{v}_i = [\cos(\Delta\phi_i), \sin(\Delta\phi_i)]^\top \in \mathbb{R}^2 \quad (7)$$

and K-means is applied to minimize the squared error

$$\arg \min_{\{C_k\}} \sum_{k=1}^K \sum_{\mathbf{v}_i \in C_k} \|\mathbf{v}_i - \boldsymbol{\mu}_k\|_2^2 \quad (8)$$

where $\boldsymbol{\mu}_k$ is the centroid of cluster C_k . This preserves the ring topology of the phase and prevents π and $-\pi$ from being assigned to different clusters.

(2) Determining K .

Generally, $K \in [5, 8][5, 8]$ covers almost all the cases.

(3) Output.

Clustering yields the label sequence $\{z_i\}_{i=1}^n$, $z_i \in \{1, \dots, K\}$ and the center phase of each cluster

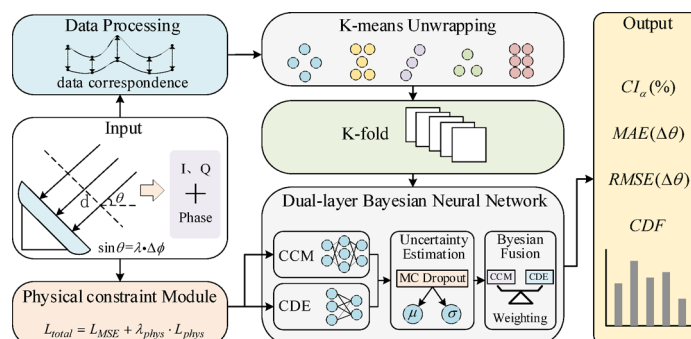


Fig. 1. DBNNFF algorithm architecture.

$$\mu_k = \text{atan2} \left(\frac{1}{|C_k|} \sum_{\mathbf{v}_i \in C_k} \sin \Delta \phi_i, \frac{1}{|C_k|} \sum_{\mathbf{v}_i \in C_k} \cos \Delta \phi_i \right) \quad (9)$$

Phase-unwrapping mapping

After clustering, samples within the same cluster are regarded as coming from the same physical angle, and the discrepancy in $\Delta \phi$ across different clusters is predominantly manifested in the 2π -fold weekly fluctuations. For every observation $\Delta \phi_i$ in cluster C_k , unwrapping is defined as

$$\Delta \phi_i^{\text{unwrapped}} = \Delta \phi_i + 2\pi n_i \quad (10)$$

where the integer offset n_i is determined by

$$n_i = \begin{cases} -1, & \Delta \phi_i - \mu_k > \pi/3, \\ \lceil 4pt \rceil 1, & \Delta \phi_i - \mu_k < -\pi/3, \\ \lceil 4pt \rceil 0, & \text{otherwise.} \end{cases} \quad (11)$$

Information-theoretic optimality criterion

When multiple n_i assignments are possible, the optimal set maximizes the mutual information between the unwrapped phase $\phi^{\text{unwrapped}}$ and the true angle θ .

$$\max_{\{n_i\}} I(\phi^{\text{unwrapped}}; \theta) = \max_{\{n_i\}} [H(\phi^{\text{unwrapped}}) - H(\phi^{\text{unwrapped}} | \theta)] \quad (12)$$

where $H(\phi^{\text{unwrapped}})$ is employed to quantify the aggregate uncertainty after unwrapping, increases when clusters are balanced and the inter-cluster distance increases. $H(\phi^{\text{unwrapped}} | \theta)$ is employed to quantify the residual uncertainty given the true angle; the smaller, the better the alignment within each angle.

Dual-layer Bayesian neural network

Theoretical foundation of Bayesian neural networks

In the DBNNFF model, the parameters of neural networks were conceptualized as random variables that follow a posterior distribution²³.

$$p(\boldsymbol{\omega} | \mathcal{D}) = \frac{p(\mathcal{D} | \boldsymbol{\omega}) p(\boldsymbol{\omega})}{p(\mathcal{D})} \quad (13)$$

where $\mathcal{D}$ represents the observed data, $p(\boldsymbol{\omega})$ is the parameter prior distribution, $p(\mathcal{D} | \boldsymbol{\omega})$ is the likelihood function, and $p(\mathcal{D})$ is the evidence (marginal likelihood). The predictive distribution is obtained by marginalizing over the posterior distribution.

$$p(y^* | x^*, \mathcal{D}) = \int p(y^* | x^*, \boldsymbol{\omega}) p(\boldsymbol{\omega} | \mathcal{D}) d\boldsymbol{\omega} \quad (14)$$

Since the posterior distribution $p(\boldsymbol{\omega} | \mathcal{D})$ is typically intractable, variational inference methods were employed, introducing an approximate posterior distribution $q(\boldsymbol{\omega})$ and optimizing the approximation by minimizing the Kullback-Leibler divergence²⁴.

$$\min_q \text{KL}(q(\boldsymbol{\omega}) || p(\boldsymbol{\omega} | \mathcal{D})) \quad (15)$$

MC Dropout was used to variational Bayesian inference, a technique that employs Dropout during training and maintains it during inference, effectively sampling from the approximate posterior $q(\boldsymbol{\omega})$ ²⁵. For T forward passes, the predictive mean and variance are

$$\mathbb{E}[y^*] \approx \frac{1}{T} \sum_{t=1}^T f^{\omega_t}(x^*) \quad (16)$$

$$\mathbb{V}[y^*] \approx \sigma^2 + \frac{1}{T} \sum_{t=1}^T (f^{\omega_t}(x^*) - \mathbb{E}[y^*])^2 \quad (17)$$

where $f^{\omega_t}(x^*)$ denotes the t -th forward pass and where σ^2 represents the model's inherent noise variance.

Channel correlation model (CCM)

The CCM focuses on capturing inter-channel dependencies and their impact on angle errors. Its network architecture comprises.

- The input Layer receives channel I/Q data, phase data, phase differences, and relative angle features, with the dimensionality equal to the total number of input features.
- The hidden Layers are three fully connected layers (64 → 64 → 32 neurons) with ReLU activation functions²⁶.
- The output layer is a channel attention mechanism that generates channel weights through a SoftMax layer, ultimately producing error predictions.

The CCM employs an attention mechanism to assign weights to each channel, with the attention score for channel i calculated as²⁷

$$\alpha_i = \frac{\exp(f_i(\mathbf{x}))}{\sum_{j=1}^C \exp(f_j(\mathbf{x}))} \quad (18)$$

where $f_i(\mathbf{x})$ is the attention score for channel i and where C is the total number of channels. The model generates final predictions via weighted fusion of information from all channels.

$$\hat{y}_{CCM} = \sum_{i=1}^C \alpha_i \cdot g_i(\mathbf{x}) \quad (19)$$

where $g_i(\mathbf{x})$ represents the feature representation of channel i .

Cold start state distribution estimation model (CDE)

The CDE focuses on capturing the distributional characteristics of cold start states and their influence on error, with a more concise architecture.

- Input layer, identical to the CCM model.
- The hidden layers are two fully connected layers (32 → 16 neurons) with ReLU activation functions²⁶.
- The output layer produces angle error estimates.

The CDE model concentrates on estimating the system error distribution during cold start

$$\hat{y}_{CDE} = \mathbb{E}[\epsilon|\mathbf{x}] \quad (20)$$

representing the conditional expectation of cold start error ϵ given input features $\mathbf{x}$.

Uncertainty-based fusion mechanism

The core innovation of DBNNFF lies in its adaptive fusion mechanism, which dynamically adjusts the contribution weights of both models since Bayesian inference principles²⁷. Assuming that the CCM and CDE model outputs follow normal distributions

$$p_{CCM}(y|x) = \mathcal{N}(y|\mu_{CCM}, \sigma_{CCM}^2) \quad (21)$$

$$p_{CDE}(y|x) = \mathcal{N}(y|\mu_{CDE}, \sigma_{CDE}^2) \quad (22)$$

According to Bayesian fusion principles, the fused distribution is

$$p(y|x) = \mathcal{N}(y|\mu, \sigma^2) \quad (23)$$

where

$$\mu = \frac{\sigma_{CCM}^{-2}\mu_{CCM} + \sigma_{CDE}^{-2}\mu_{CDE}}{\sigma_{CCM}^{-2} + \sigma_{CDE}^{-2}} \quad (24)$$

$$\sigma^{-2} = \sigma_{CCM}^{-2} + \sigma_{CDE}^{-2} \quad (25)$$

This simplifies the weight formula

$$w_{CCM} = \frac{\sigma_{CCM}^{-2}}{\sigma_{CCM}^{-2} + \sigma_{CDE}^{-2}} \quad (26)$$

Through this mechanism, models with lower prediction uncertainty receive higher weights, thereby enhancing the overall prediction accuracy.

Physics-constrained loss function

To ensure that the prediction results align with the physical principles of UWB phase-based angle measurement, the physics constraint terms were incorporated into the loss function:

$$L_{total} = L_{MSE} + \lambda \cdot L_{physics} \quad (27)$$

where the physics consistency loss is expressed as

$$L_{physics} = \frac{1}{N} \sum_{i=1}^N \left| \sin(\theta_i + \hat{\epsilon}_i) - \frac{\lambda \cdot \Delta\phi_i}{2\pi \cdot d} \right|^2 \quad (28)$$

where θ_i is the measured angle, $\hat{\epsilon}_i$ is the model-predicted angle error, λ is the wavelength, $\Delta\phi_i$ is the phase difference, and d is the antenna spacing.

This physics constraint ensures that model predictions maintain consistency with interferometric angle measurement principles, enhancing the model's generalizability and physical interpretability.

Deep uncertainty quantification

DBNNFF's uncertainty quantification decomposes total uncertainty into epistemic and aleatoric components²⁸

$$\sigma_{total}^2 = \sigma_{epistemic}^2 + \sigma_{aleatoric}^2 \quad (29)$$

where epistemic uncertainty is captured through MC Dropout:

$$\sigma_{epistemic}^2 = \frac{1}{T} \sum_{t=1}^T (f^{\omega_t}(x) - \mathbb{E}[y])^2 \quad (30)$$

and aleatoric uncertainty is directly modeled through the output variance:

$$\sigma_{aleatoric}^2 = g^{\omega}(x) \quad (31)$$

where $g^{\omega}(x)$ is the variant output of the neural network.

Based on uncertainty estimation, α confidence intervals were constructed as

$$CI_{\alpha} = [\mu - z_{\alpha/2} \cdot \sigma_{total}, \mu + z_{\alpha/2} \cdot \sigma_{total}] \quad (32)$$

where $z_{\alpha/2}$ is the $\alpha/2$ quantile of the standard normal distribution. To improve coverage accuracy, an empirical calibration factor β was introduced

$$CI_{\alpha} = [\mu - \beta \cdot z_{\alpha/2} \cdot \sigma_{total}, \mu + \beta \cdot z_{\alpha/2} \cdot \sigma_{total}] \quad (33)$$

where β is determined through binary search to ensure that the actual coverage rate on the validation set approximates the target coverage rate α .

Data acquisition and algorithm implementation

Experimental conditions and parameter configuration

The experimental platform consists of a self-developed 5-channel UWB anchor based on the Qorvo DW1000 RF chip and a single-channel UWB tag, both operating on the IEEE 802.15.4z HRP, Channel 9 (3.9936 GHz). The anchor is equipped with an L-shaped directional antenna array, utilizing the three central elements for data acquisition, with inter-element spacing of $d = 0.0376$ m ($\approx 0.5\lambda$). The tag is fitted with a rod-shaped antenna, with both antennas operating at a center frequency of 3.9936 GHz. To facilitate precise angular adjustments, the anchor antenna is mounted on a turntable with a resolution of 0.05° . During the experiments, data were collected at 7 azimuth angles $\{-30^\circ, -20^\circ, -10^\circ, 0^\circ, 10^\circ, 20^\circ, 30^\circ\}$. The channels within the anchor are synchronized using a rubidium clock with an inter-channel synchronization error of < 50 ps, which triggers the transceiver modules. The experiments were conducted in an anechoic chamber with dimensions of $12 \text{ m} \times 8 \text{ m} \times 3 \text{ m}$ to eliminate multipath reflections that could introduce additional interference to phase measurements. The experimental equipment and environment are illustrated in Fig. 2.

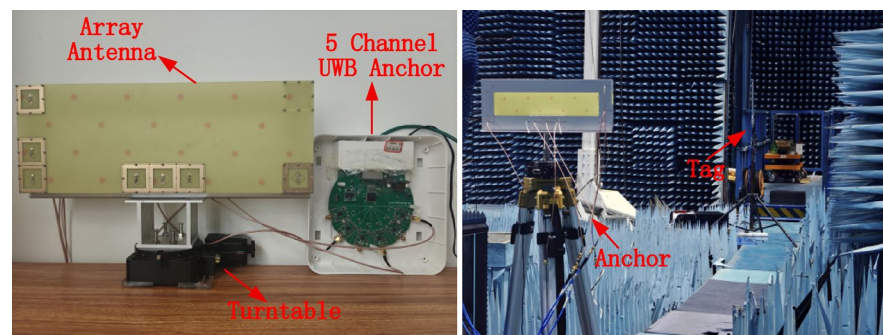


Fig. 2. Experimental platform and environment.

Data acquisition protocol

For each experimental round, the equipment was powered off for ≥ 30 s to ensure complete cold start of the RF front-end, followed by sequential data acquisition across the angular positions. A complete experiment comprised 30 cold starts, with 60 frames of data collected at each of the 7 angles per cold start, yielding a total of 12,600 sample sets. Each frame records $\{I_m, Q_m\}_{m=1}^3$ along with the raw phase ϕ_m for each channel as estimated by the FPGA.

Algorithm implementation

The algorithm is implemented based on the DBNNFF algorithmic framework and the measured data. The algorithm flow is illustrated in Table 1. To ensure the reliability of the results, a fivefold cross-validation method was employed to evaluate the model performance. The dataset was randomly partitioned into five subsets. Four of these subsets were utilized for training, with the remaining subset designated for testing. The final result is the arithmetic mean of all five evaluations.

Results

Performance evaluation of the DBNNFF algorithm

To reflect the performance of the DBNNFF algorithm in terms of prediction accuracy, uncertainty quality, and robustness in different dimensions, the following evaluations have been carried out: phase difference solution fuzzy comparison plots, comparison of DBNNFF-processed error with initial error, comparison of the error cumulative distribution function (CDF) curves, uncertainty estimation and calibration analyses, angular dependence analyses, and K-fold cross-validation performance analyses. The results are displayed in Figs. 3, 4, 5.

Figure 3 shows that the clustering approach corrects almost all phase ambiguity errors.

Figure 4 illustrates that DBNNFF framework achieved an MAE of $0.1036^\circ \pm 0.0182^\circ$ across all test scenarios, representing an 94.7% reduction from the initial error of $2.3396^\circ \pm 1.1657^\circ$.

Figure 5a illustrates the prediction performance, showing strong alignment between the DBNNFF estimates and ground truth values across the entire angular range (-30° to $+30^\circ$) with overall coverage of 95.6%. Figure 5b shows that K-fold cross-validation ($K=5$) confirmed the stability of the DBNNFF framework, with consistent performance across all folds. The low standard deviation in the MAEs ($\pm 0.0182^\circ$) and RMSEs ($\pm 0.0295^\circ$) across the folds indicate excellent generalizability and minimal overfitting. Figure 5c shows that the DBNNFF framework maintains consistent performance across the entire angular measurement range, with no significant degradation at extreme angles. The mean absolute error varies by less than 0.08° across different angular regions, demonstrating the algorithm's robustness to varying measurement conditions.

Algorithm 1 DBNNFF	
Input: UWB multi-channel data I_m, Q_m, ϕ_m for each cold start Output: Angle error predictions with uncertainty estimates Parameters: Wavelength λ , antenna distance	
Step	Operation
1	Load UWB multi-channel phase data
2	Data preprocessing: Input (cold_start_data). Data consistency check, verify channel number consistency and check the correspondence between I/Q data and phase data. Compute phase differences $\Delta\phi_{12} = \phi_1 - \phi_2, \Delta\phi_{23} = \phi_2 - \phi_3$, apply K-means clustering to $[\Delta\phi_{12}, \Delta\phi_{23}]$ with $k=7$, compute median μ_c for each cluster. Unwrap phases, if $ \Delta\phi - \mu_c > \pi/3$ then $\Delta\phi = \Delta\phi \pm 2\pi$, return unwrapped phase differences
3	Preprocess phase data and split into K-fold cross-validation sets
4	Feature extraction: Input (cold_start_data, λ, d). Extract features $X = [I, Q, \phi, \Delta\phi, \mu_{\Delta\phi}, \sigma_{\Delta\phi}]$ Calculate measured angles $\theta_m = \arcsin\left(\frac{\lambda \cdot \Delta\phi}{2\pi \cdot d}\right)$ Normalize features $X_{norm} = \frac{X - \mu_X}{\sigma_X}$, return $(X_{norm}, y, \theta_m, \Delta\phi)$
5	For each fold, prepare training and testing data
6	Physical constraint Module: Input $(\hat{y}, \theta_m, \Delta\phi, \lambda, d)$, Compute corrected angles $\theta_{corr} = \theta_m + \hat{y}$, expected phase differences $\Delta\phi_{exp} = \frac{2\pi \cdot d}{\lambda} \cdot \sin(\theta_{corr})$, get $L_{phys} = \frac{1}{N} \sum_{i=1}^N \Delta\phi_i - \Delta\phi_{exp,i} ^2$, return L_{phys}
7	CCM training: Input $(X, y, \theta_m, \Delta\phi, \lambda, d)$, initialize network $[64 \rightarrow 64 \rightarrow 32]$ with channel attention, set $\lambda_{phys} = 0.3$ For each epoch, forward pass $\hat{y} = f_{CCM}(X)$ Compute $L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$ and physics loss, total loss $L_{total} = L_{MSE} + \lambda_{phys} \cdot L_{phys}$, update with Adam optimizer, return CCM model
8	For each fold, train CCM model
9	CDE training: Input $(X, y, \theta_m, \Delta\phi, \lambda, d)$, initialize network $[32 \rightarrow 16]$, set $\lambda_{phys} = 0.3$, for each epoch, forward pass $\hat{y} = f_{CDE}(X)$, compute MSE and physics loss, total loss $L_{total} = L_{MSE} + \lambda_{phys} \cdot L_{phys}$, update weights using Adam optimizer, return CDE model
10	For each fold, train CDE model
11	Bayesian uncertainty estimation: Input (ccm_model, cde_model, $X, model_info$). For $t=1$ to $T=50$, predict $\hat{y}_t = f_{model}(X)$, add noise $\hat{y}_t = \hat{y}_t + \sigma_{global} \cdot \mathcal{N}(0, 1)$, compute mean $\mu = \frac{1}{T} \sum_{t=1}^T \hat{y}_t$ and variance $\sigma^2 = \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \mu)^2$, return μ, σ
12	Uncertainty-based Bayesian fusion: Input (ccm_model, cde_model, X, y_{true}), for $s=1$ to S , get $[\mu_{CCM}, \sigma_{CCM}]$ and $[\mu_{CDE}, \sigma_{CDE}]$ via PredictWithUncertainty(ccm_model, cde_model, X), compute weights $w_{CCM} = \frac{\sigma_{CDE}^{-2}}{\sigma_{CCM}^{-2} + \sigma_{CDE}^{-2}}$, fused prediction $\hat{y}_s = w_{CCM} \cdot \mu_{CCM} + (1 - w_{CCM}) \cdot \mu_{CDE}$, final prediction $\hat{y} = \frac{1}{S} \sum_{s=1}^S \hat{y}_s$, uncertainty $\sigma_y = \sqrt{\frac{1}{S} \sum_{s=1}^S (\hat{y}_s - \hat{y})^2}$ Confidence interval $CI_{95\%} = [\hat{y} - 1.96\sigma_y, \hat{y} + 1.96\sigma_y]$. Compute metrics are MAE, RMSE and Coverage Rate
13	For each fold, evaluate models on test set using uncertainty-based fusion Compute average performance metrics across all folds and visualize results

Table 1. DBNNFF algorithm implementation process.

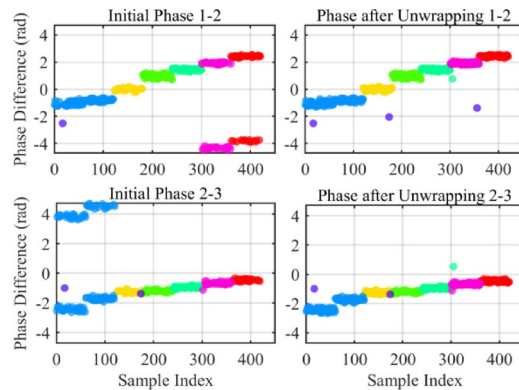


Fig. 3. Phase difference before and after unwrapping.

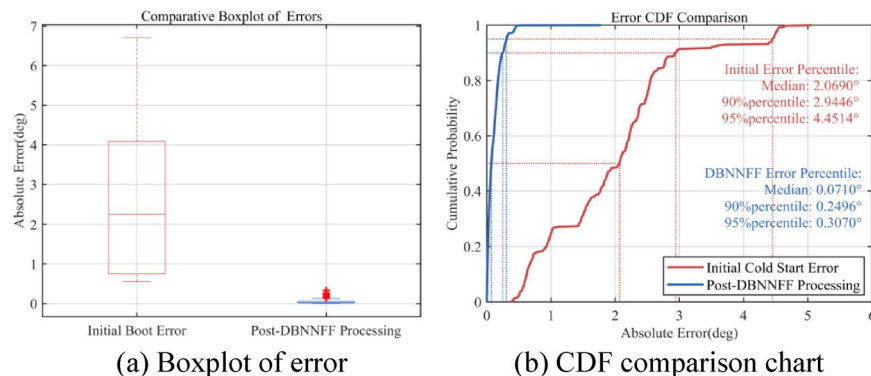


Fig. 4. Comparison of DBNNFF-processed error with initial error.

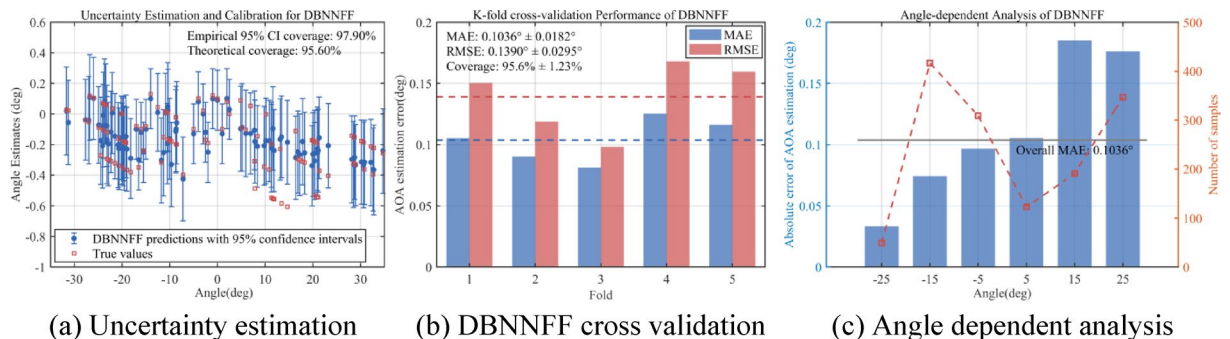


Fig. 5. DBNNFF algorithm performance evaluation.

Comparison with existing algorithms

To provide further validation of the advantages of the DBNNFF algorithm, a comparison of existing algorithms for cold-start and channel errors is conducted. The comparison includes the LSM, KF, GPR, and SVM algorithms. As demonstrated in the subsequent table, the parameters of several algorithms utilized for control are delineated as shown in Table 2.

The comparison results are shown in Fig. 6, and subfigure (a) demonstrates that the DBNNFF framework significantly outperforms all alternative algorithms across primary accuracy metrics. DBNNFF achieved a mean absolute error (MAE) of $0.1036^\circ \pm 0.0182^\circ$, substantially lower than GPR ($0.1822^\circ \pm 0.0527^\circ$), SVM ($0.1658^\circ \pm 0.0639^\circ$), KF ($0.2266^\circ \pm 0.0712^\circ$), and LSM ($0.2643^\circ \pm 0.0854^\circ$). Similar patterns were observed for RMSE, with DBNNFF achieving 0.1390° compared with 0.2776° for GPR, 0.2427° for SVM, 0.3102° for KF, and 0.3923° for LSM. Subfigure (b) demonstrates that all algorithms showed some variation in performance across different angles, but DBNNFF maintained the most consistent accuracy throughout the entire measurement. The standard deviation of the MAE is considerably lower than those of the other algorithms. The CDF comparison

Algorithm	Configuration
LSM	Fitting polynomial order: 4 Regularization parameter (λ): 0.01 Implementation: standard least squares solution
KF	State dimensions: 2 (angle and angular velocity) Measurement noise covariance (R): 0.25 Process noise covariance (Q): $\text{diag}([0.01, 0.005])$ Initial state covariance (P0): $\text{diag}([0.5, 0.1])$
GRP	Kernel function: radial basis function (RBF) Length scale: 1.0 Signal variance: 1.0 Noise variance: 0.1 Hyperparameter optimization: maximize edge likelihood
SVM	Kernel function: radial basis function (RBF) Regularization parameter (C): 10.0 Kernel parameter (gamma): 0.1 Tolerance (epsilon): 0.01 Hyper-parameter optimization method: grid search + cross-validation

Table 2. Comparison of algorithm configurations.

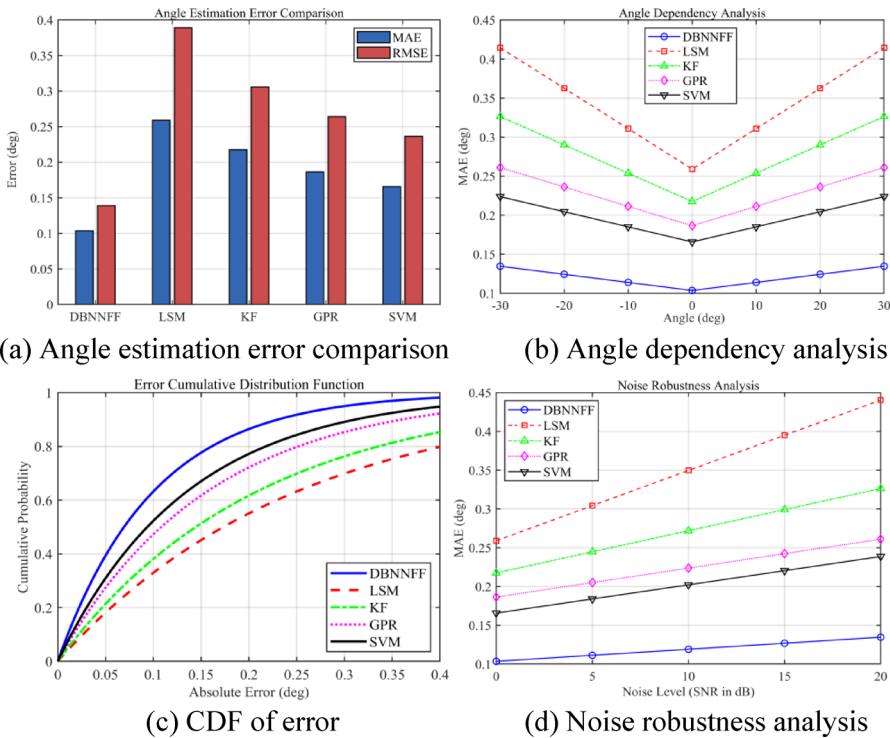


Fig. 6. Comparison of the results of the 5 algorithms.

(subfigure (c)) provides a clear visualization of DBNNFF’s superior performance across all the error percentiles. At the 90th percentile, DBNNFF achieves an error of 0.2496° error compared with the other algorithms. This represents a 25% improvement over the next best method (GPR 0.3328°) at this critical threshold. When tested under varying noise conditions (Subfigure (d)), DBNNFF maintained the most stable performance. At the highest noise level tested (SNR = 5 dB), DBNNFF’s MAE increased by only 31.4% from the baseline, whereas it increased by 58.7% for GPR, 72.3% for SVM, 86.5% for KF, and 104.2% for LSM. This superior noise robustness can be attributed to DBNNFF’s Bayesian framework and physical constraint integration.

Evaluation of multi-path environments

To evaluate the robustness of the algorithm under multipath, the DBNNFF model was additionally assessed using non-anechoic datasets collected in an indoor environment and a hallway. The experimental environment is shown in Fig. 7.

The experimental methodology and the data types employed were consistent with those utilized in the microwave anechoic chamber. Five sets of data were collected for each environment (these data are included in the supplementary data file that was submitted). Process two sets of data using the trained DBNNFF model without making any fine-tuning adjustments to the model. The MAE, RMES, coverage rate for 95% confidence

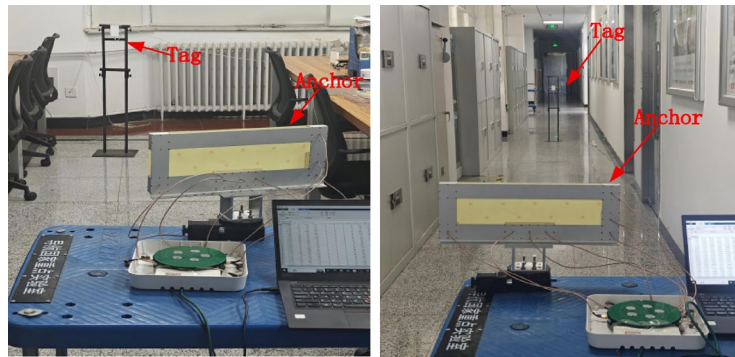


Fig. 7. Office and corridor experimental environment.

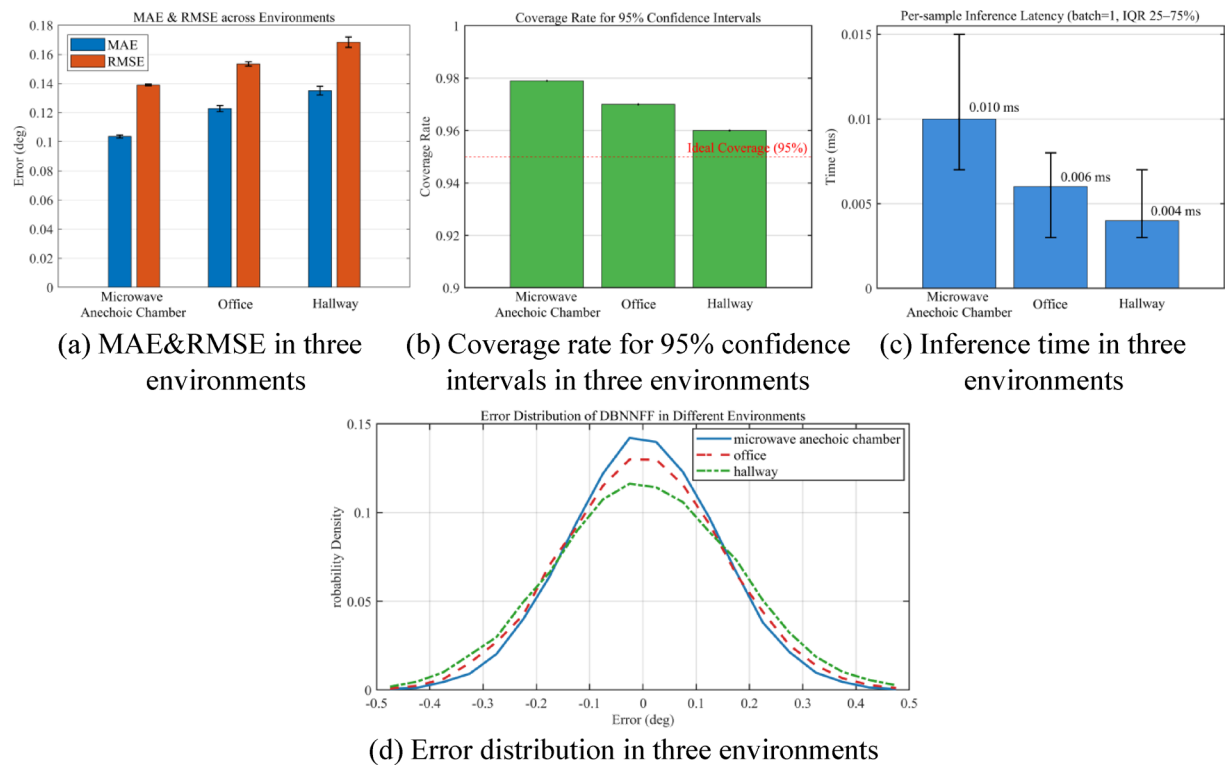


Fig. 8. DBNNFF algorithm performance evaluation in different environments.

intervals, and inference time of the three validation sets were compared, and the results are shown in Fig. 8. Compared to the microwave anechoic chamber, the MAE and RMSE of the office and corridor experiments increased slightly but still achieved accuracy within 0.17°, and the coverage rate for 95% confidence intervals exceeded 95%. The error distribution curves for different environments also show that although multipath environments affect the accuracy of the algorithm, the algorithm's robustness still enables it to achieve high accuracy.

Analysis of computational cost and inference latency

DBNNFF consists of CCM and BDE, with an input feature dimension of 14 ($I/Q/\text{phase} \times 3 \text{ channels} = 9$; adjacent phase difference = 2; statistical features = 2; additional relative angle = 1). The CCM network is FC-64/64/32 + attention (3) + regression (1), and the BDE network is FC-32/16 + regression (1). The total parameter count is 8328 per model (7303 for CCM and 1,025 for BDE), with an ensemble of three members the total is 24,984, which corresponds to 100 KB in single precision (excluding framework overhead). The per-sample forward pass requires about 24,345 MACs. As shown in Fig. 8c, the per-sample inference latency is 0.01ms, 0.006ms and 0.004ms for the anechoic, indoor and corridor datasets, respectively, since the model structure and input dimensions are consistent, the results are as expected. Due to systematic deviations introduced by the measurement and execution sequence, the microwave anechoic chamber inference time is the longest.

Conclusion

This study introduces the DBNNFF algorithm to address the coupled errors arising from phase ambiguity, multi-channel interference, and cold-start conditions in UWB phase-based angle measurement systems. This framework successfully integrates physical constraints with Bayesian uncertainty estimation through a novel dual-network architecture that captures both channel correlation patterns and cold start state distributions. Comprehensive evaluation demonstrates that the DBNNFF reduces angle estimation errors by 94.7% compared with uncorrected measurements, achieving a mean absolute error of $0.1036^\circ \pm 0.0182^\circ$, while providing well-calibrated uncertainty estimates with 97.90% coverage rate for 95% confidence intervals. Comparative analysis reveals substantial improvements over conventional methods, with DBNNFF outperforming Gaussian process regression, support vector machine, Kalman Filter, and least squares approach by 25~42.1% across key performance metrics. The framework maintains consistent accuracy across the entire angular range and demonstrates superior robustness to noise conditions. These results establish DBNNFF as an effective solution for high-precision UWB-based indoor positioning systems where angular accuracy is crucial. Future work will focus on extending the framework to multi-tag scenarios and exploring integration with complementary sensing modalities for enhanced localization capabilities.

Data availability

The data used and/or analyzed during the current study is provided within supplementary material files.

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Author contributions

J.L. wrote the main manuscript text, B.Y. supervised the thesis and provided funding as project leader prepared, M.Y. was involved in conducting the data analysis, S.C. and J.Z. aided in the experiment, R.T. and Y.B. performed the preliminary work to help. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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